MATTHEW TRENT

SOFTWARE DEVELOPER

CONTACT

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- **All my links**: matthewtrent.me/socials

EDUCATION

- Honors Computer Science, University of Victoria, BSc, with R&D project
- 4.1 GPA
- Near full-ride academic scholarships

HACKATHONS

- **UVIC Engineering '23**: 1st place; tutor app
- **UVIC Engineering '24**: 2nd place; real-time cloud-first collaborative code editor in 6 hours
- **Island Health Code Hack**: Recognized for discovering serious security vulnerability
- **UVIC AI**: Developed competitive Monte Carlo tree-search bot
- **Covehack '24**: Founded, organized, and judged

SOCIALS

- **Coding Instagrams**: ~76,000 followers
- **Coding Discord**: ~1,400 members
- **Stack Overflow**: ~3,200 reputation
- **Tech blog**: 9 articles and 5 series

PUBLIC PACKAGES

- **qc:** Homebrew quantum computing circuit designer and simulator
- **jsonencryption**: Go package for managing JSON field encryption
- **use_optimistic**: My port of React.js' useOptimistic hook to Flutter
- **killswitch**: Flutter app remote kill switch
- **shrinking_view**: Flutter widget x/y/z-axis transform manipulation
- **alert_banner**: Custom alert banner for Flutter apps
- **scrollable**: Flutter widget manager
- **core_utils**: Solves common Flutter issues
- **ghloc**: Homebrew GitHub commit tracker

PUBLIC TOOLS

- **Real-time Markdown badge**: ~33,000 uses
- **GitHub graph painter**: ~500 script uses
- **Leetcode bot**: Solved 1000s of questions
- **Discord bot**: Moderating 1000+ users
- **Online school bot**: Does class for users
- **Poll bot**: Overwhelms & controls Menti polls

WHO I AM

I'm an extremely enthusiastic developer excited to use my skills and knowledge to create innovative and effective solutions for a wide range of technical problems. I have a strong work ethic and am always eager to learn and improve my abilities.

PROFESSIONAL EXPERIENCE

CORTEX Labs • Software Developer

Sept. 2023 - June 2024

- Productionized the DECISION-MAP (database of evidence concerning interventions supporting the intersections of brain injury mental health and addictions problems) website.
- Developed this **big-data visualization tool** that aggregated clinical studies relating to brain injuries in Vue.js (Typescript).
- Built in association with the **dean of research.**

Van Belle Nursery • Software Developer

Summer 2020

- Championed the adoption of Flutter to **company leadership** for internal use.
- Developed a POC cross-platform mobile application for use across business verticals: Time logging, finances, authentication, data table storage/searching, etc.

NOTABLE PROJECTS

Decentralized Peer-to-Peer Search Engine (R&D)

April 2024 - Present

- Allows for node-based **partitioning of a decentralized search index governed by utility crypto token**.
- Maintains trustless infrastructure via **Go-based P2P networking**, **ZKPs**, **consensus algorithms for persisting trusted public keys**, and the maintenance of general state via multiple **Kademlia-based distributed hash tables**.
- Supports semantic query matching as nodes each store replicas/shards of a maintained **384-dimensional vector embedding space**.
- Boasts a backlink, page-content, and general popularity combinational search ranking algorithm **powered by independent network crawler nodes**.
- In development.

University-Exclusive Social Media App

March 2022 - Present

- Fullstack cross-platform social media application. Server and client utilize tech such as Dart, Docker, GCP, AWS, Go, PostgreSQL, Redis, Firebase, Javascript, Flutter, etc.
- Implements: **Infinite feeds, threaded comments, trending/recent/sentiment-based content sorting, live chat, notification systems, custom authentication, hottest daily posts, achievements, profiles, curated content by individual campus, admin/mod roles**, and more.
- Large project. I've led a team of 5 developers on this, managing tasks, PRs, GitHub Issues, etc. It contains **hundreds-of-thousands of lines of code**.

iMessage Clone

May 2023

- Created a functional version of Apple's iMessage in Flutter and Rust.
- Utilizes custom **Rust Actor-based websocket protocol** for real-time communication.
- Boasts features like: **Message backup, reactions, deleting/editing messages, blocking users, authentication flow, "is active now" indicators**, etc.
- Turned out to be a fair use-case for a non-relational database.

Real-Time Predictor of Hockey Match Outcomes

April 2024

- Helped create a real-time hockey game outcome predictor using **logistic regression** and **LSTM neural networks**.
- Leveraged over **1.75 million shot entries** from MoneyPuck with 42 engineered features, data preprocessing, and predictive modeling for enhanced prediction accuracy.